

International Institute of Information Technology, Hyderabad
CS7.404: Digital Image Processing
Assignment 0

Last date of Submission: *Friday, 14 Aug, 2026; 11pm*

The goal of this assignment is to get you familiarised with reading, writing and basic manipulation of images and videos. You are expected NOT to use any image libraries for this assignment, except for reading and writing from a file.

Upload should contain a pdf file describing how you implemented each function, any interesting learnings, and output images if any. In addition, you should also upload a zip file containing your complete code.

Implement the following functions:

1. Read an image file into an array
2. Write an array into an image file
Note: Make sure you can handle color and grayscale images
3. Change the brightness of the image.
4. Change the contrast of the image
5. Change a color image to grayscale.
Qn: Are there different ways of doing this? What is the visual effect of each?
6. Convert a grayscale image to color using a pseudo color mapping.
7. Read an image with green screen for background and read another image which has a different background. The function should replace the green screen with the corresponding pixels of the background present in the second image.
8. Read a video file and convert it into a sequence (array) of images and also write it back as a video.
9. Create a 1 second transition video (fade, slide, any other) from one image to another.